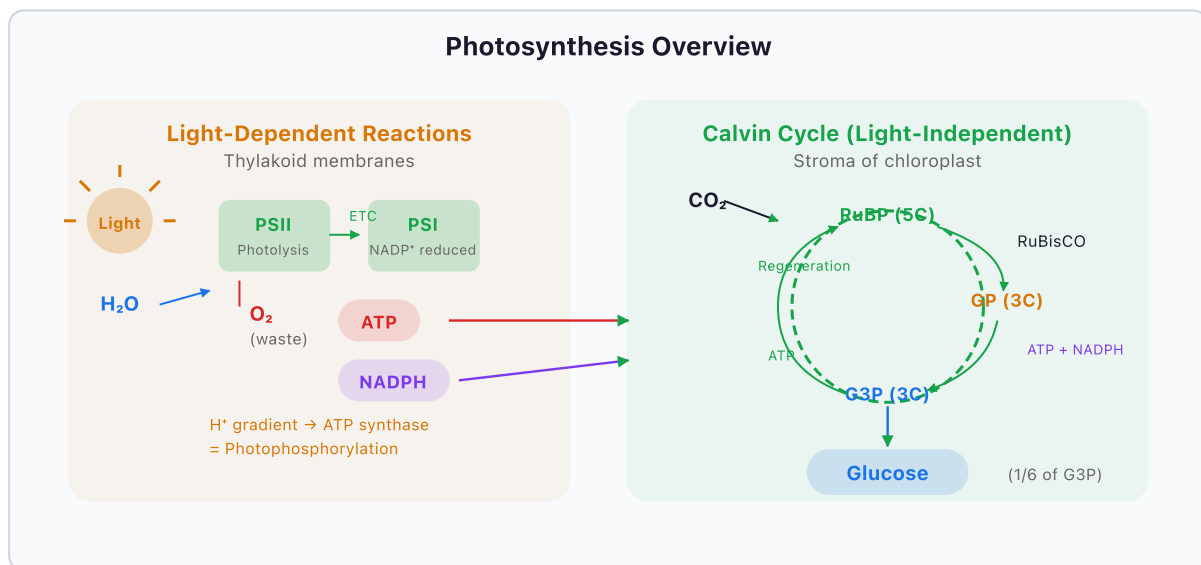


S5 — On the Wild Side

5.1 Photosynthesis

Overview of Photosynthesis

Photosynthesis: The process by which plants, algae, and some bacteria convert light energy into chemical energy (stored in glucose), using carbon dioxide and water as raw materials and releasing oxygen as a by-product.



Photosynthesis occurs in two stages: the light-dependent reactions (in the thylakoid membranes) and the light-independent reactions (in the stroma of chloroplasts).

Overall, photosynthesis uses energy from light to split apart the strong bonds in water molecules (photolysis). The hydrogen is stored in a fuel (glucose) by combining it with carbon dioxide, and the oxygen is released into the atmosphere.

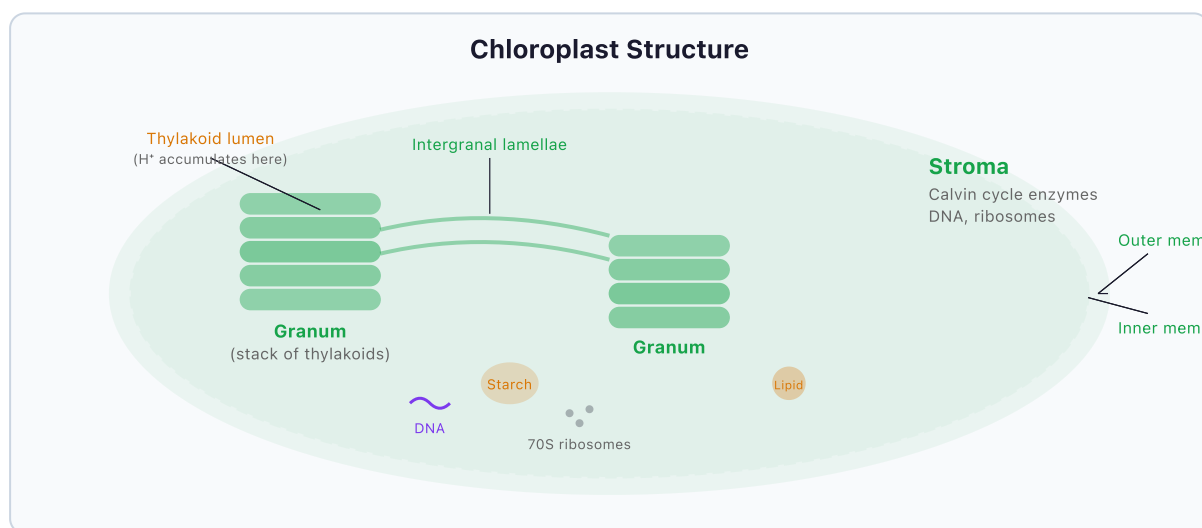
ATP: Phosphorylation and Hydrolysis

ATP (adenosine triphosphate) is the immediate energy source for biological processes.

Phosphorylation of ADP ($\text{ADP} + \text{P}_i \rightarrow \text{ATP}$) requires energy, supplied in photosynthesis by light (photophosphorylation) and in respiration by the oxidation of glucose. **Hydrolysis** of ATP ($\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{P}_i$), catalysed by ATPase, releases energy in small, manageable amounts immediately wherever it is needed, e.g. for active transport, muscle contraction and the synthesis of large molecules.

Chloroplast Structure

Chloroplasts are surrounded by a double membrane (envelope). Internally, they contain an extensive system of thylakoid membranes arranged in stacks called grana (singular: granum), connected by intergranal lamellae. The thylakoid membranes contain photosynthetic pigments (chlorophyll a, chlorophyll b, carotenoids), electron carriers, and ATP synthase enzymes. The stroma is the fluid-filled matrix surrounding the grana, containing enzymes for the Calvin cycle, DNA, ribosomes, starch grains, and lipid droplets.



Light-Dependent Reactions

The light-dependent reactions occur on the thylakoid membranes and require light energy. The key events are:

1. **Photoexcitation:** Light energy is absorbed by chlorophyll molecules in photosystem II (PSII). Electrons in chlorophyll are excited to a higher energy level and are captured by electron acceptors.
2. **Photolysis of water:** The lost electrons are replaced by splitting water molecules (photolysis): $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$. This is the source of the oxygen released in photosynthesis.

3. **Electron transport chain:** Excited electrons pass along a chain of electron carriers in the thylakoid membrane, losing energy at each step. This energy is used to pump H^+ ions from the stroma into the thylakoid space, creating a proton gradient.
4. **Chemiosmosis:** H^+ ions flow back to the stroma through ATP synthase (down the electrochemical gradient), driving the synthesis of ATP from ADP + P_i . This is called photophosphorylation.
5. **Photosystem I (PSI):** Electrons from the ETC are re-energised by light in PSI and transferred to $NADP^+$ reductase, which combines them with H^+ ions to reduce $NADP^+$ to NADPH (reduced NADP).

The products of the light-dependent reactions are ATP, NADPH (both used in the Calvin cycle), and O_2 (released as a waste product).

Light-Independent Reactions (Calvin Cycle)

The Calvin cycle occurs in the stroma and does not directly require light, but it depends on the ATP and NADPH produced by the light-dependent reactions. The key steps are:

1. **Carbon fixation:** CO_2 combines with the 5-carbon acceptor molecule ribulose biphosphate (RuBP), catalysed by the enzyme RuBisCO. This produces an unstable 6-carbon intermediate that immediately splits into two molecules of 3-carbon glycerate-3-phosphate (GP).
2. **Reduction:** GP is reduced to glyceraldehyde-3-phosphate (G3P, also called GALP or triose phosphate) using ATP and NADPH from the light reactions. ATP provides energy; NADPH provides reducing power (hydrogen).
3. **Regeneration of RuBP:** Most G3P molecules (5 out of every 6) are used to regenerate RuBP using ATP. This keeps the cycle running. One in six G3P molecules is the net product, used to synthesise glucose, amino acids, lipids, or other organic molecules.

Exam Tip: For every 3 CO_2 molecules that enter the Calvin cycle, 6 G3P are produced but 5 are recycled to regenerate 3 RuBP. Only 1 G3P is the net gain. It takes 6 turns of the cycle (6 CO_2) to produce one glucose molecule (6 carbons = 2 × G3P combined).

The simple sugars made in the Calvin cycle are used by plants, animals and other organisms in **respiration** and in the **synthesis of new biological molecules**: polysaccharides (e.g. starch, cellulose), amino acids (with nitrate from the soil), lipids and nucleic acids.

Core Practical 11: Investigate photosynthesis using isolated chloroplasts (the Hill reaction).

Leaves are blended in ice-cold, isotonic buffer and the extract is filtered and centrifuged to isolate chloroplasts (cold slows enzyme activity; isotonic prevents osmotic damage; buffer keeps pH constant). A blue redox dye such as DCPIP is added: it accepts electrons from the light-dependent reactions in place of NADP and turns colourless when reduced. The rate of decolourisation (measured with a colorimeter) shows the rate of the light-dependent reactions. Controls: chloroplasts kept in the dark, and dye with no chloroplasts.

5.2 Ecosystems and Energy Transfer

Ecosystem Concepts

Ecosystem: A community of organisms (biotic component) interacting with each other and with their physical environment (abiotic component) in a defined area. Ecosystems are characterised by energy flow and nutrient cycling.

Key terms: a **habitat** is the place where an organism lives; a **niche** is the role of an organism within its ecosystem (what it eats, when it is active, where it lives, how it interacts with other species); a **population** is all the individuals of one species in a habitat; a **community** is all the populations of different species in a habitat.

The numbers and distribution of organisms in a habitat are controlled by **abiotic factors** (non-living: e.g. light intensity, temperature, water availability, soil pH, mineral ions, oxygen) and **biotic factors** (living: e.g. competition, predation, grazing, parasitism, disease). The **niche** concept accounts for distribution and abundance: each species can only survive where conditions match its niche, and two species cannot occupy exactly the same niche in a habitat for long, because the better-adapted one outcompetes the other (competitive exclusion). Species with similar requirements therefore partition resources or occupy different parts of the habitat.

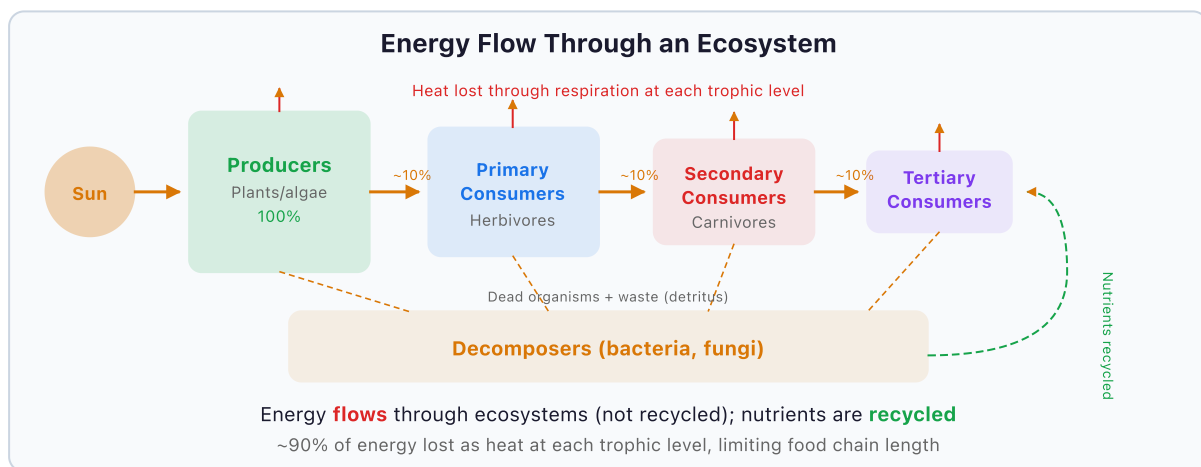
Core Practical 10: Ecology of a habitat. Use random quadrats (coordinates from random number tables) to estimate abundance (percentage cover, frequency or density) in a uniform area, or a belt or line transect with quadrats at regular intervals to study distribution along an environmental gradient (e.g. a sand dune or woodland edge). Measure appropriate abiotic factors at each quadrat

(e.g. light meter, soil thermometer, soil moisture and pH, wind speed) and look for correlations, e.g. using Spearman's rank correlation coefficient.

Energy Flow Through Ecosystems

Energy enters most ecosystems as sunlight and is fixed by producers (autotrophs) through photosynthesis. Energy flows through the ecosystem via food chains and food webs: producers → primary consumers → secondary consumers → tertiary consumers. At each trophic level, energy is lost through:

- **Respiration:** The largest loss. Energy is released as heat during metabolic processes and cannot be recycled.
- **Excretion and egestion:** Not all ingested material is assimilated. Faeces contain undigested material; urine contains nitrogenous waste.
- **Death:** Dead organisms and waste are broken down by decomposers (bacteria and fungi), releasing energy as heat.



Typically only about 10% of energy is transferred from one trophic level to the next (though this varies: endotherms lose more energy through respiration for thermoregulation than ectotherms, so transfer efficiency is often lower in mammalian food chains). This limits food chains to usually 4-5 trophic levels and explains why there is less biomass at higher trophic levels (ecological pyramids of energy are always upright). It also explains why there are fewer top predators than herbivores in any ecosystem, and why food chains in aquatic ecosystems can sometimes be longer (ectothermic organisms lose less energy as heat).

Real-world application: The low efficiency of energy transfer between trophic levels has significant implications for food security. Producing 1 kg of beef requires approximately 8–10 kg of grain feed, because most of the energy in the grain is lost through the cow's respiration, movement, and metabolic processes. This is why eating lower on the food chain (more plant-based diets) is more energy-efficient and can support larger human populations. Intensive farming practices aim to increase energy transfer efficiency by restricting animal movement (reducing respiratory losses) and controlling temperature (reducing thermoregulation costs), though these raise animal welfare concerns.

$$\text{Efficiency of energy transfer} = (\text{energy available at next trophic level} / \text{energy available at previous trophic level}) \times 100\%$$

Gross and Net Primary Productivity

$$\text{NPP} = \text{GPP} - \text{R}$$

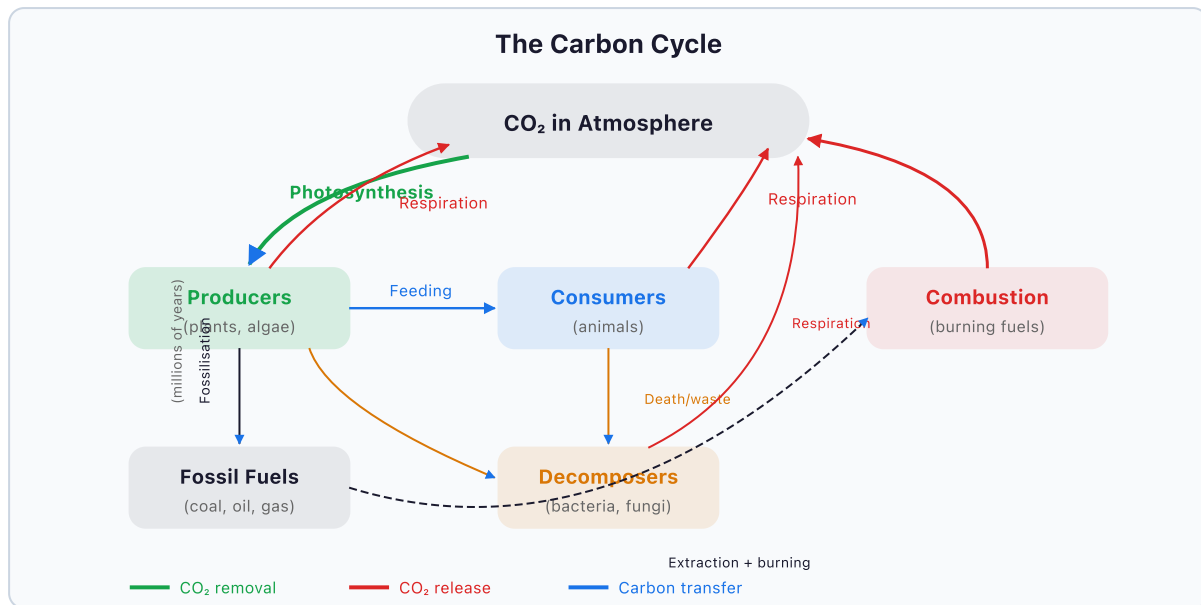
Gross primary productivity (GPP) is the total rate at which energy is incorporated into organic molecules by producers through photosynthesis. Net primary productivity (NPP) is the rate at which energy is stored in plant biomass after respiratory losses are subtracted. NPP represents the energy available to primary consumers.

Calculating NPP: if a grassland has a GPP of $25\,000 \text{ kJ m}^{-2} \text{ year}^{-1}$ and plant respiration uses $15\,000 \text{ kJ m}^{-2} \text{ year}^{-1}$, $\text{NPP} = 25\,000 - 15\,000 = 10\,000 \text{ kJ m}^{-2} \text{ year}^{-1}$. If primary consumers take in $1\,000 \text{ kJ m}^{-2} \text{ year}^{-1}$ of this, the efficiency of transfer from producers is $(1\,000 / 10\,000) \times 100 = 10\%$. Always give productivity with units of energy (or biomass) per unit area per unit time.

5.3 Climate Change

The Greenhouse Effect

The greenhouse effect is a natural phenomenon that maintains Earth's temperature at a habitable level. Short-wave radiation from the Sun passes through the atmosphere and warms the Earth's surface. The surface re-emits longer-wave infrared radiation, some of which is absorbed by greenhouse gases (CO_2 , methane, water vapour, nitrous oxide) and re-radiated back to Earth, warming the atmosphere.



The **enhanced greenhouse effect** refers to the additional warming caused by increased concentrations of greenhouse gases due to human activities, primarily:

- **Burning of fossil fuels:** Releases CO₂ that was locked away in geological carbon sinks for millions of years, adding net carbon to the active carbon cycle. Atmospheric CO₂ has risen from ~280 ppm pre-industrial to over 420 ppm.
- **Deforestation:** Reduces CO₂ absorption by photosynthesis and releases stored carbon when trees are burned or decompose. Tropical deforestation accounts for ~10% of global emissions.
- **Agriculture:** Methane (CH₄) from enteric fermentation in ruminant livestock and anaerobic conditions in rice paddies. Methane is ~80 times more potent as a greenhouse gas than CO₂ over 20 years, though it persists in the atmosphere for a shorter time (~12 years vs centuries for CO₂).
- **Nitrous oxide (N₂O):** From agricultural fertilisers and industrial processes. ~300 times more potent than CO₂ per molecule.

Positive feedback loops may accelerate climate change: warming melts permafrost, releasing trapped methane; reduced ice cover decreases albedo (reflectivity), so more solar radiation is absorbed; warmer oceans absorb less CO₂ (gases are less soluble in warm water). These feedback mechanisms mean that initial warming can trigger further warming, potentially leading to tipping points beyond which change becomes self-reinforcing and irreversible.

Evidence for Climate Change

- **Ice core data:** Air bubbles trapped in ice cores from Antarctica and Greenland provide a record of atmospheric CO₂ and temperature going back hundreds of thousands of years. There is a strong correlation between CO₂ concentration and temperature.
- **Temperature records:** Global average temperatures have risen by approximately 1.1°C since pre-industrial times, with the rate of warming accelerating since the mid-20th century.
- **Sea level rise:** Thermal expansion of seawater and melting of ice caps and glaciers.
- **Retreating glaciers and sea ice:** Satellite measurements show significant losses.
- **Phenological changes:** Earlier flowering, earlier migration, shifts in species ranges toward the poles or higher altitudes.
- **Dendrochronology:** Tree ring analysis showing changes in growth patterns.
- **Pollen in peat bogs:** Changes in plant communities over time.

How the proxy records work: in **dendrochronology**, trees lay down one ring a year and rings are wider in warm, wet years, so ring widths (dated by counting back) indicate past climate. In **peat bogs**, acidic, anaerobic, waterlogged conditions stop pollen decaying; pollen grains can be identified to plant species, and since each species has known climatic requirements, the pollen in each dated layer shows the past climate. **Correlation and causation:** records show a correlation between rising CO₂ and rising temperature, but a correlation alone does not prove that one causes the other; a causal link needs a plausible mechanism (the greenhouse effect) and supporting experimental evidence.

Models of Future Climate Change

Data on CO₂ levels and temperature can be **extrapolated** beyond the range of measurements to make predictions, and these are used in computer models of future climate change under different emissions scenarios. Models have **limitations**: extrapolation assumes current trends continue; our understanding of the climate system is incomplete (e.g. feedbacks from clouds, oceans, ice and permafrost); future human emissions are uncertain; and past data are limited in accuracy. Different models therefore give a range of predictions.

Consequences of Climate Change

Climate change affects biodiversity through habitat loss (coral bleaching, sea level rise), disruption of ecological relationships (mismatched timing between species, e.g., caterpillars emerging before birds breed), range shifts, increased extreme weather events, and ocean acidification (CO₂ dissolving to form carbonic acid, affecting shell-forming organisms).

For plants and animals the effects of changing rainfall patterns and seasonal cycles include changes in the **distribution of species** (ranges shift towards the poles or to higher altitude as conditions become unsuitable) and changes to **development and life cycles** (e.g. earlier leafing and flowering, earlier egg-laying and insect emergence, altered hatching or sex ratios in reptiles whose sex is set by incubation temperature).

Exam Tip: Distinguish between the natural greenhouse effect (essential for life) and the enhanced greenhouse effect (caused by human activity). When discussing evidence, always evaluate its strengths and limitations (e.g., correlation does not prove causation, but the mechanism is well understood).

Temperature and Enzyme Activity

Rising temperature increases the kinetic energy of enzyme and substrate molecules, so there are more successful collisions and more enzyme–substrate complexes form, and the rate increases up to the optimum. Above the optimum, bonds holding the tertiary structure break, the active site changes shape and the enzyme is **denatured**, so the rate falls. Because all organisms depend on enzymes, warming affects plants (e.g. rates of photosynthesis, germination and growth), animals (especially ectotherms, whose body temperature follows the environment) and microorganisms (e.g. faster decomposition releasing more CO₂).

$$Q_{10} = \text{rate at } (t + 10)^{\circ}\text{C} / \text{rate at } t^{\circ}\text{C}$$

The temperature coefficient Q_{10} is the factor by which the rate increases for a 10°C rise; for most enzyme-controlled reactions between about 0 and 40°C it is approximately 2.

Core Practical 12: Investigate the effect of temperature on the initial rate of an enzyme-catalysed reaction, to include Q_{10} (e.g. trypsin digesting milk protein, followed with a colorimeter, in water baths at a range of temperatures). Find the initial rate from the gradient of a tangent at the start of the curve, then calculate Q_{10} . **Core Practical 13: Investigate the effects of temperature on the development of organisms** (e.g. seedling growth rate or brine shrimp hatch rates) at a range of controlled temperatures, keeping other variables constant.

Controversy, the Carbon Cycle and Conservation

Who reaches the conclusions: scientific conclusions about controversial issues, such as how far humans are affecting climate change or what actions should be taken, can depend on who is reaching them. Scientists, governments, industries (e.g. fossil-fuel companies), environmental pressure groups and the media may interpret the same data differently because of funding, vested interests or their values, so evidence should be judged on its source, methods and whether it has been peer reviewed.

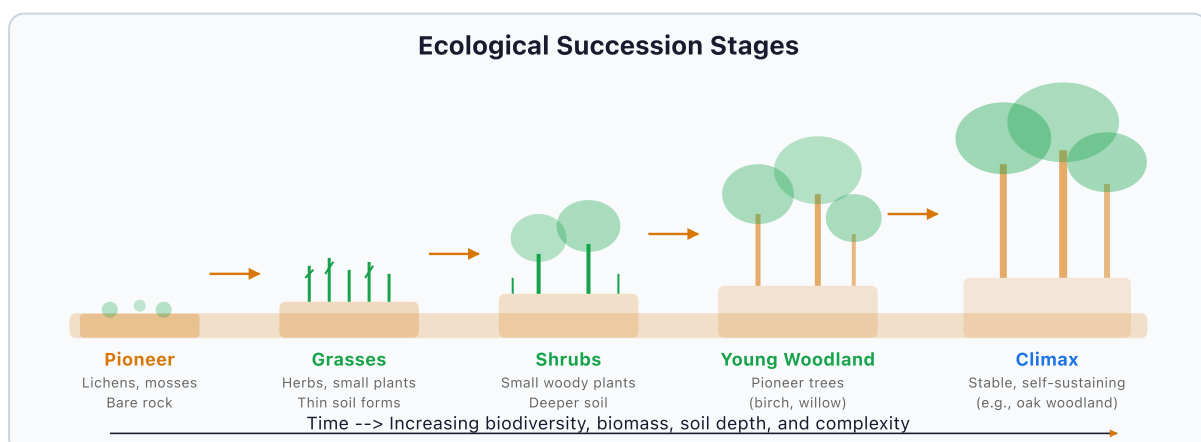
Using the carbon cycle to reduce atmospheric CO₂: increase uptake by photosynthesis (reforestation and afforestation, protecting existing forests and peat bogs as carbon sinks) and reduce release by combustion and respiration (burning less fossil fuel, using renewable energy and biofuels).

Reforestation and sustainable resources manage the conflict between human needs and conservation. Replanting forests (and harvesting timber sustainably, replacing trees that are felled) provides wood and jobs while restoring habitats and removing CO₂. **Biofuels** (e.g. bioethanol, biodiesel) are close to carbon neutral, since the CO₂ released on burning was recently absorbed in photosynthesis, but growing biofuel crops can compete with food crops for land and may drive deforestation.

5.4 Ecological Succession

Succession

Succession: The gradual and directional change in the species composition of a community over time, eventually leading to a stable climax community. Primary succession begins on bare, previously uncolonised substrate; secondary succession occurs after disturbance of an existing community.



Pioneer species (e.g., lichens, mosses) are the first to colonise bare rock. They are adapted to harsh conditions and help create soil by weathering rock and adding organic matter when they die. As soil develops, more complex plant communities can establish, each seral stage modifying the environment to make it more suitable for the next stage. Eventually, a climax community is reached (determined by climate and soil type), which is relatively stable and self-sustaining.

5.5 Evolution and Speciation

Natural Selection

Natural selection: The mechanism of evolution by which organisms with advantageous traits are more likely to survive, reproduce, and pass their alleles to the next generation. Over time, the frequency of beneficial alleles increases in the population.

The conditions for natural selection are: genetic variation within a population (caused by mutation, meiosis, and sexual reproduction); competition for limited resources (overproduction of offspring); differential survival and reproduction (organisms with favourable traits are more likely to survive); and inheritance (advantageous traits are passed to offspring).

Types of Selection

- **Directional selection:** Individuals at one extreme of a phenotypic range have greater fitness. The population mean shifts in one direction. Example: peppered moths during industrial pollution (darker moths had higher survival).
- **Stabilising selection:** Individuals with intermediate phenotypes have greatest fitness; extremes are selected against. Reduces variation but does not change the mean. Example: human birth weight (very small or very large babies have lower survival).
- **Disruptive selection:** Individuals at both extremes have greater fitness than those in the middle. Can lead to speciation. Example: beak size in seed-eating birds on islands with only very large or very small seeds.

Speciation

Speciation: The formation of new species from an existing population, occurring when populations become reproductively isolated so that gene flow ceases, and they diverge genetically until they can no longer interbreed to produce fertile offspring.

Allopatric speciation: Geographical isolation separates a population (e.g., mountain range, river, island colonisation). The isolated populations experience different selection pressures and genetic drift, accumulating genetic differences until they become reproductively incompatible.

Sympatric speciation: New species arise within the same geographical area without physical barriers. Mechanisms include polyploidy (especially in plants, where chromosome doubling creates instant reproductive isolation), ecological specialisation, or temporal isolation (breeding at different times).

Evidence for Evolution

Evidence includes the fossil record (showing change over time and transitional forms), comparative anatomy (homologous structures indicating common ancestry), molecular biology (DNA and protein sequence similarities), biogeography (distribution of species reflecting evolutionary history), and direct observation of evolution (antibiotic resistance in bacteria, pesticide resistance in insects).

The role of the scientific community: new evidence for evolution, including **genomics** (comparing DNA base sequences) and **proteomics** (comparing amino acid sequences of proteins), is validated before it is accepted. Scientists publish their work in **scientific journals** only after **peer review**, in which other experts in the field check the methods, data and conclusions; work is also presented and discussed at **scientific conferences**. This allows others to repeat the work and critically evaluate it. The more closely related two species are, the more similar their DNA and protein sequences, which supports the accepted theory of evolution.

Exam Tip: When describing natural selection, use precise language: variation exists, those with advantageous traits are more likely to survive and reproduce, these traits are inherited, and allele frequency changes over generations. Avoid Lamarckian language (organisms do not "choose" to evolve or "develop" traits because they need them).